06 · RAP Research — Data Verification & How Commitments Vary

This document explains **two things we learned while building the platform** that shape how the data can and cannot be used:

1. **How we verify data sources** — the discipline that stops the AI from inventing facts, and how every figure on the dashboard traces back to a real, public document.
2. **How Reconciliation Action Plans (RAPs) actually vary** — in the words they use and, more importantly, in the *quality* of their commitments: many carry no due date and no measurable target, which has direct consequences for what a tracking platform can honestly report.

It closes with **how the platform is designed to cope with both**, and an honest account of where the limits are.

Companion documents: 01 · Project Audit, 02 · Deploy Runbook, 03 · Engine Comparison, 04 · Monitoring & Security, 05 · Design Decisions & User Journeys. Unfamiliar terms (Bedrock, Lambda, BN, …) are defined in the glossary in the README.

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0 · What this document is based on (read this first)

Everything below is grounded in one of three kinds of evidence, and it matters which is which:

- **The data model and extraction rules** — how the software is *built* to handle RAPs. This is strong, first-hand evidence: the design was shaped, deliberately, around messy real documents.
- **Real spot-checks** — a small set of real, human-verified material: the **Bank of Canada RAP** (our one hand-verified "gold" set), a set of **hand-curated real figures** drawn from primary public sources (`real-fixtures.ts`), and the **live n=8 engine comparison** (see doc 03).
- **The curated dashboard dataset** — the ~100 organizations populating the running dashboard are **real Canadian organizations**, and their commitments were **hand-curated from those organizations' own public disclosures** (reconciliation / ESG pages, first-party news releases), with **every source link audited** (102 of 102 resolve; see A.8). They were **not** produced by running the AI extraction pipeline on RAP PDFs, and the figures are *illustrative snapshots* from the cited sources (confirm against the source before quoting a number as exact). Separately, **3 fictional demo company accounts** contribute 9 self-submitted commitments purely to demonstrate the self-report flow — those are the only fabricated records.

We are careful throughout to say *"the model and our real spot-checks confirm this,"* not *"we measured this across a large real corpus."* We have **not** run the extraction pipeline across a large body of real RAPs — that requires the client's own AWS account (see doc 02). The observations here are nonetheless well-grounded: where we make a claim about real RAPs, we point to a real example.

Part A · How we verify data sources

The platform holds two very different kinds of data, and "verification" means something different for each. Keeping them separate is the first honesty rule.

A.1 · Two data regimes

- **The curated public-disclosure dataset** (behind the RAP Index dashboard). Each record is drawn from an **organization's own public disclosure** — its reconciliation/ESG page or a first-party news release — and stores the source link. This is deliberately **not** sensitive Indigenous community data.
- **The AI extraction pipeline** (uploading a RAP PDF and reading commitments out of it). This is where the risk of AI "hallucination" lives, and where most of the verification machinery is aimed.

A.2 · The core rule: the AI must *quote*, or the value is thrown away

The single most important design decision is this: **the AI's job is to locate and quote, never to know or compute.** Every field the model extracts is wrapped in a structure that requires a **verbatim quote** and a **page number** from the source document. If the model cannot produce the exact words it read a value from, **the value is discarded** (set to null) rather than kept.

In plain terms: the system would rather report *"we didn't find a due date"* than **invent** one. A confident, made-up citation is treated as worse than an honest gap.

A.3 · A deterministic check proves the quote is really in the document

A quote alone isn't enough — a model can fabricate a quote too. So after extraction, a piece of **ordinary, non-AI code** checks that the quoted text **actually appears in the document the model was shown**. It matches on words (so harmless whitespace or OCR punctuation differences don't cause false alarms) but still catches fabricated or "welded-together" quotes.

This check earned its place. During development it caught a real, dangerous bug: on two-column pages, an earlier extraction produced **21 of 32 quotes that were fabricated** — plausible-looking citations stitched from fragments, with off-by-one page numbers. The verbatim-quote gate is what now stops that class of error. On the validated run it produced **zero false positives across ~180 quotes**.

A.4 · A human confirms before anything becomes official

Nothing an AI extracts reaches the live dashboard automatically. Every extraction lands in a **Pending Review** state; a human reviewer must confirm it (field by field) before it becomes a canonical record. And any number that matters — a target value, a due date, a percentage — is **parsed and calculated in code, never by the language model.** The AI finds the words; the software does the arithmetic.

A.5 · Sources are ranked honestly: confirmed > research > self-reported

Not all data deserves equal trust, and the platform never pretends otherwise. Each record carries a **basis**:

- **Confirmed** — independently corroborated (e.g. procurement dollars attested by the supplier on the other side of the transaction). Only these can raise a commitment to "confirmed."
- **Research** — curated by our team from an organization's public disclosure. This is the default for the seed dataset.
- **Self-reported** — entered by a company about itself. These are **opt-in** (surfaced only when the organization's identity is registry-verified and it opts in) and **never count toward the headline figures** — opting in makes them *shown*, not *counted*.

A self-report can **never** be silently promoted to "confirmed." Identity itself is anchored to the organization's **CRA Business Number** (with a checksum pre-filter and a registry lookup), so records attach to a real legal entity rather than a name that might be spelled three different ways.

A.6 · The engine choice was measured, not guessed

Which AI reading engine to trust was settled by a **live, billed comparison across 8 documents** (full detail in doc 03). The recommendation — **Textract-LAYOUT** — won because it produced **real page numbers that were read from the document** (not inferred), solid quote grounding, and was the only engine to process all 8 documents without crashing. A cheaper text-layer engine is the residency-friendly fallback.

A.7 · We proved that "let a second AI check the first" does not work here

A tempting shortcut is to have a second AI judge whether the first AI's findings are correct. **We tested this and it failed.** Two independent judge models agreed with each other essentially only by chance (statistical agreement $\kappa \approx \mathbf{0}$) — because both simply assented to almost everything. On real, unlabeled RAP data, AI cross-checking gives **no reliable signal**. When humans adjudicated the handful of genuine disagreements, the split fell exactly on the *vague-and-inferred* vs *specific-and-grounded* line — which is precisely why a **human**, not an automated AI check, is the right reviewer. This is why the human-in-the-loop gate in A.4 is not bureaucratic caution; it is the only instrument that actually works.

A.8 · The curated dataset is link-audited

For the public-disclosure dataset, **every source URL was HTTP-checked** — 102 of 102 unique links resolve to a live first-party page, with no fabricated URLs and no status ever inflated beyond what the source supports. Figures in that dataset are **illustrative snapshots** taken from the cited sources; the standing instruction is to confirm against the source before quoting a number as exact.

A.9 · Honest limitations of the verification story

A client should know these plainly:

- **AI inference leaves Canada.** Data can rest in Canada, but Amazon's Bedrock models are not hosted in Canada, so the reading step routes to a US/global region regardless of engine. (See doc 03; a Canadian-hosted model such as TELUS's would close this gap.)
- **Our one human "gold" reference is a single document** (Bank of Canada). Broader accuracy is measured *relative to the combined output of all engines* — a defect every engine misses is invisible.
- **The production default engine (BDA) infers page numbers** rather than reading them, and is brittle on unusual PDFs. A back-fill step recovers verbatim quotes — and reliable pages

— for the values it can locate in the document's own text layer (best-effort, PDF-only; any value it can't locate keeps BDA's inferred page). Textract-LAYOUT is the recommendation precisely to avoid depending on that back-fill.
- **Scanned / image-only PDFs are untested** — the corpus was all "born-digital," and the text-layer engine has no OCR.
- **The business-registry (ISED) integration is coded but not yet activated** against the live API; today it runs as a stub.

Part B · How RAPs actually vary

Reconciliation Action Plans are **not** a standardized filing. They vary enormously — first in vocabulary, and more consequentially in the *substance* of what they commit to. This is the single biggest challenge for any platform that tries to track them, and it is worth the client understanding it directly.

B.1 · They don't even agree on what the document is called

Across real organizations, the same kind of artifact goes by different names:

- **Bank of Canada** — "Reconciliation Action Plan"
- **TELUS** — "Indigenous Reconciliation & Connectivity Report"
- **Suncor** — "Report on Sustainability — Indigenous Relations"
- **TC Energy** — "Indigenous Relations & Reconciliation"
- **Government of Canada** — "Mandatory Minimum 5% Indigenous Procurement Target"

The internal structure varies just as much: Bank of Canada organizes commitments into **"pathways"** ("People pathway," "Learning pathway"); OPG uses a **5-pillar** structure; the Australian *Reflect RAP* format (Populous) mandates a rigid **"Action / Deliverable / Timeline / Responsibility"** table with maturity tiers (*reflect / innovate / stretch / elevate*) that have **no Canadian equivalent at all**.

How the platform copes: every commitment stores **both** the document's own wording *and* a normalized value mapped onto a single canonical vocabulary (18 sectors, a fixed set of commitment types and themes). Nothing is lost — the original words are always kept beside the standardized one, so the dashboard can compare across organizations without erasing how each one actually spoke.

Be precise about how this works: the mapping from idiosyncratic wording to the canonical vocabulary is done by the **AI at extraction time and confirmed by the human reviewer** — it is not a fixed dictionary of synonyms. So terminology unification, like everything else, ultimately rests on the human-in-the-loop review gate.

B.2 · The commitments themselves vary in structure

The extraction data model is built, deliberately, to **absorb** this variation rather than force every RAP into one shape:

- **Only two things are effectively required** of a commitment: an **action** and a **deliverable**.
- **Everything else is optional** and frequently absent: the **owner**, the **timeline / due date**, and any **quantitative target**.

- Sector-specific fields are filled in only when relevant, and a genuinely novel field surfaces in an "extras" bucket rather than being force-fit or invented — **absence is treated as meaningful information**, not an error.

B.3 · Many commitments have no due date

This is real and common. A great many RAP commitments are phrased as an **ongoing cadence** ("Annual," "Ongoing," "Every three years") or carry **no timeline at all**. Real examples from our verified material:

- **TELUS** — *"Maintain a public Indigenous reconciliation action plan"* (no date)
- **Agnico Eagle** — *"Publish a Reconciliation Action Plan"* (no date)
- The **Bank of Canada gold set** (our human-verified transcription) is stored with no dates, because the document states none.

These are governance- or announcement-style statements. Read strictly, a commitment with **no date and no measure is not trackable** — there is no moment at which it can be said to be met or missed. From an accountability standpoint, an undated "maintain / publish / continue" pledge does little to demonstrate progress, however sincere the intent behind it. We note this as a **data-quality and accountability observation**, grounded in the documents themselves — not an accusation of motive.

(Design note: the platform stores the verbatim timeline wording next to the parsed date precisely so that a real cadence like "Annual" isn't silently destroyed into "no timeline." An earlier version made that mistake.)

B.4 · Many commitments have no measurable target

Alongside missing dates, many commitments carry **no metric, KPI, or numeric target** — they are purely qualitative. The Bank of Canada set is explicitly **"qualitative pathways with no headline dollar targets,"** and the RAP-publication pledges above (TELUS, Agnico) name no measure; these undated, unmeasurable commitments read as bare action verbs: *"Develop a framework…," "Engage in regular dialogue…," "Seek guidance from…," "Maintain…"* (The point is not that these organizations have no metrics — TELUS and Agnico do report hard dollar figures on *other* commitments — but that a substantial share of individual commitments carry nothing measurable at all.)

A commitment with neither a target nor a date offers a tracking platform **nothing to track**. This is the crux of the accountability gap: the plan can be published and celebrated without ever exposing a checkpoint against which follow-through could be judged.

B.5 · An honest caveat about these two patterns

The platform **records** whether a date or target is present (an empty value beside the preserved verbatim text), but it does **not yet flag or score** undated / unmeasurable commitments as a quality problem. In other words: the data to surface "how many of this organization's commitments are actually trackable?" exists, but there is currently **no feature that presents it that way**. That is an honest gap — and, as Part C notes, a natural next step.

Part C · How the platform is designed to address this

Pulling Parts A and B together, here is how the product responds to messy sources and uneven commitments today, and where it can go next.

Already built:

- **Normalization that never erases the original.** Canonical vocabulary for cross-organization comparison, with each document's own words retained alongside (B.1).
- **Grounding + human review** that absorbs structural variation safely: nullable-by-contract fields mean a RAP that omits owners, dates, or targets is handled honestly rather than hallucinated into completeness (A.2–A.4, B.2).
- **A progress and risk engine.** Commitments carry a status (committed / in-progress / reported / confirmed / stalled) and an append-only history; the platform computes **overdue** and **at-risk** flags and produces plain-language digests ("N commitments are past their target year without confirmation").
- **A confirmation-integrity view.** The dashboard measures the gap between what is *self-reported* and what is *independently confirmed* — the very gap the platform exists to surface — and never inflates a self-report into a confirmation.

Natural next step (not yet built):

- **A "trackability" signal.** Because the platform already records whether each commitment has a date and a measurable target, it could surface, per organization, *how many of its commitments are actually trackable* versus undated/unmeasurable — turning the observation in B.3–B.4 into a first-class, at-a-glance accountability indicator. This would be a modest, high-value addition built entirely on data the system already captures.

Part D · Evidence base & references

Real, verified material (safe to rely on):

- **Bank of Canada gold set** — `scripts/fixtures/gold-commitments-bankofcanada.json` (one human-verified RAP; the source of the "qualitative, undated" examples).
- **Hand-curated real figures** — `src/lib/rap/real-fixtures.ts` (real public figures drawn from primary sources by the team; real-world-true, but hand-curated, *not* produced by running the extraction pipeline).
- **Live engine comparison** — `docs/rap-engine-comparison.md` (billed n=8 run; the $\kappa \approx 0$ finding and the Textract-LAYOUT recommendation).
- **Extraction findings** — `docs/rap-extraction-findings.md` (the fabricated-quote bug, the verbatim-quote gate, over-extraction fixes).
- **Curated-dataset verification** — `DATA_VERIFICATION.md` (the 102/102 URL audit and the status-integrity rules).
- **Real RAP corpus (11 documents / ~250 pages)** — inventory in `Week 7/rap_samples/README.md`. These are readable source PDFs; extracting them at scale requires the client's AWS account (doc 02), as the local demo returns canned data.

Where the mechanisms live in code (for a technical reader):

- The locate-and-quote contract and the canonical/optional field model — `src/lib/rap/types.ts`.
- The extraction rules (including "normalize themes" and "forward-looking only") — `src/lib/rap/extraction-schema.ts`.
- The verbatim-quote gate — `src/lib/rap/validate.ts`.
- Source ranking (confirmed > research > self-reported) — `src/lib/index-evidence/resolver.ts`, `status-map.ts`.
- The overdue / at-risk risk engine — `src/lib/commitments/insights.ts`.

Honesty summary for the client: the *design* evidence is strong and first-hand; the *real-RAP* evidence is small but genuine (one gold set + curated real figures + a live 8-document run). The organizations populating the running dashboard are **real, hand-curated from public disclosures** (only 3 demo login accounts are fictional). What we have **not** done is run the AI extraction pipeline across a large body of real RAP *PDFs* — so the fine-grained, commitment-level observations in Part B (undated / unmeasurable commitments) rest on the extraction data model plus real spot-checks (the Bank of Canada gold set and curated real figures), not on a large extracted corpus. Every claim in Parts A and B about real RAPs is backed by a real example above; none of it should be read as "measured across a large real corpus."

Point-in-time handoff document. The living technical sources are in the repository at the paths cited above; the engine comparison (doc 03) is a live n=8 billed run, and the $\kappa \approx 0$ result reproduces a published "ecological boundary" finding — LLM cross-checking gives no decorrelated verification signal on this kind of real, unlabeled data, which is why human review is retained by design.